



/apispec_1.json

Explore

User Service API 1.0.0

[Base URL: localhost:5001/]
[/apispec_1.json](#)

API for managing users and their friendships

Schemes

HTTP

Auth



POST

/login

Log in with username and password

post_login

Parameters

Try it out

Name	Description
body * required	
object (body)	Example Value Model <pre>{ "password": "mysecurepassword", "username": "johndoe" }</pre>
	Parameter content type application/json

Responses

Response content type

application/json

Code	Description
200	Login successful
	Example Value Model

Code	Description
	<pre>{ "message": "Login successful", "user_id": 1 }</pre>
400	Missing username or password
	Example ValueModel <pre>{ "error": "username and password are required" }</pre>
401	Invalid credentials
	Example ValueModel <pre>{ "error": "Invalid credentials" }</pre>

Friends



POST	/users/ Create a new user	post_users_
Parameters		Try it out
Name	Description	
body * required object (body)	Example ValueModel <pre>{ "password": "mysecurepassword", "username": "johndoe" }</pre> Parameter content type application/json	
Responses	Response content type	application/json

Code	Description
------	-------------

201

User successfully created

Example Value Model

```
{
  "message": "User created",
  "user_id": 1
}
```

400

Missing username or password

Example Value Model

```
{
  "error": "username and password are required"
}
```

409

Username already exists

Example Value Model

```
{
  "error": "Username already exists"
}
```

GET

/users/{user_id} Get a user by ID

get_users__user_id_

Parameters

Try it out

Name

Description

user_id * requiredinteger
(path)

ID of the user to retrieve

user_id - ID of the user to retrieve

Responses

Response content type

application/json

Code

Description

200

User information

Code	Description
	Example Value Model
	<pre>{ "id": 1, "username": "johndoe" }</pre>

404

User not found

Example Value	Model
<pre>{ "error": "User not found" }</pre>	

GET**/users/{user_id}/friends/** Get a user's friends**get_users__user_id__friends_****Parameters****Try it out**

Name	Description
user_id * required	ID of the user whose friends to retrieve
integer (path)	

user_id - ID of the user whose friends to retrieve

Responses

Response content type

application/json

Code	Description
200	List of user's friends
	Example Value Model
	<pre>[{ "id": 2, "username": "janedoe" }]</pre>

404

User not found

Example Value	Model
----------------------	-------

Code	Description
------	-------------

```
{
  "error": "User not found"
}
```

POST**/users/{user_id}/friends/** Add a friend to the user**post_users__user_id__friends_****Parameters**[Try it out](#)

Name	Description
------	-------------

user_id * requiredinteger
(path)

ID of the user adding a friend

user_id - ID of the user adding a friend

body * requiredobject
(body)**Example Value** Model

```
{
  "friend_id": 2
}
```

Parameter content type

application/json

Responses**Response content type**

application/json

Code	Description
------	-------------

200

Friends are already friends

Example Value Model

```
{
  "message": "Already friends"
}
```

201

Friend successfully added

Example Value Model

```
{
  "message": "Friend added"
}
```

Code	Description
------	-------------

	}
--	---

400

Invalid request

Example Value **Model**

```
{  
  "error": "Cannot add yourself as a friend"  
}
```

404

One or both users not found

Example Value **Model**

```
{  
  "error": "User or friend not found"  
}
```

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Explore

Watch History Service API 1.0.0

[Base URL: localhost:5002/]
[/apispec_1.json](#)

API for managing user watch history

Schemes

HTTP

Watch History



GET

/watch_history/{user_id}/

Get a user's watch history

get_watch_history__user_id__

Parameters

Try it out

Name	Description
user_id * required	
integer (path)	ID of the user
user_id - ID of the user	

Responses

Response content type

application/json

Code	Description
200	List of user's watch history
Example Value Model <pre>[{ "movie_id": 1, "timestamp": "2025-05-01T10:00:00Z" }]</pre>	

Code	Description
	<pre>}]</pre>
404	User not found
	Example ValueModel <pre>{ "error": "User with ID 999 not found" }</pre>

POST	/watch_history/{user_id}/{movie_id}/	Add a movie to the user's watch history	post_watch_history__user_id__movie_id__
------	--------------------------------------	---	---

Parameters		Try it out
Name	Description	
user_id * required integer (path)	ID of the user	user_id - ID of the user
movie_id * required integer (path)	ID of the movie	movie_id - ID of the movie
Responses		Response content type
		application/json

Code	Description
201	Movie added to watch history
	Example ValueModel <pre>{ "message": "Movie 1 added to user 1's history" }</pre>
404	User or movie not found
	Example ValueModel

Code**Description**

```
{  
  "error": "User with ID 999 not found"  
}
```

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Explore

Rating Service API 1.0.0

[Base URL: localhost:5003/]
[/apispec_1.json](#)

API for managing movie ratings and reactions

Schemes

HTTP

Ratings



GET

/ratings/{user_id}/ Get a user's ratings

get_ratings__user_id__

Parameters

Try it out

Name	Description
user_id * required	
integer (path)	ID of the user
user_id - ID of the user	

Responses

Response content type application/json

Code	Description
200	List of ratings with reaction counts
Example Value Model <pre>[{ "agrees": 5, "disagrees": 2, "movie_id": 1, "score": 8 }]</pre>	

Code	Description
	<pre>}]</pre>
404	User not found

POST

/ratings/{user_id}/{movie_id}/

Rate a movie

post_ratings__user_id__movie_id__

Parameters

Try it out

Name	Description
user_id * required integer (path)	ID of the user rating the movie user_id - ID of the user rating the movie
movie_id * required integer (path)	ID of the movie being rated movie_id - ID of the movie being rated
body * required object (body)	Example ValueModel <pre>{ "score": 8 }</pre> Parameter content type application/json

Responses

Response content typeapplication/json

Code	Description
201	Movie rated successfully Example ValueModel <pre>{ "message": "Rated movie 1 with score 8" }</pre>

Code	Description
400	Invalid input
404	User or movie not found

Reactions



POST

/ratings/{user_id}/{movie_id}/reaction

Add or update a reaction to a user's rating of a movie

post_ratings__user_id__movie_id_reaction

Parameters

Try it out

Name	Description
user_id * required integer (path)	ID of the user who rated the movie user_id - ID of the user who rated the movie
movie_id * required integer (path)	ID of the movie that was rated movie_id - ID of the movie that was rated
body * required object (body)	Example Value Model <pre>{ "reaction-type": 2, "reactor_id": 2}</pre> Parameter content type application/json

Responses

Response content type

application/json

Code	Description
201	Reaction added or updated

Code	Description
	Example Value Model <pre>{ "message": "Agreed with rating" }</pre>
400	Invalid input or missing fields
404	User, movie, or rating not found

GET

/reactions/{reactor_id} /

Get all reactions made by a user

get_reactions__reactor_id_ -

Parameters

Try it out

Name	Description
reactor_id * required integer (path)	ID of the reacting user reactor_id - ID of the reacting user

Responses

Response content type application/json

Code	Description
200	List of reactions Example Value Model <pre>[{ "movie_id": 1, "reaction_type": "agree", "user_id": 3 }]</pre>
404	Reactor user not found



/apispec_1.json

Explore

Recommendation Service API 1.0.0

[Base URL: localhost:5004/]
[/apispec_1.json](#)

API for movie recommendations

Schemes

HTTP

Recommendations



GET /recommendations Get random movie recommendations **get_recommendations**

Returns a list of 5 random movie IDs that have been rated by any user

Parameters

Try it out

No parameters

Responses

Response content type

application/json

Code

Description

200

A list of randomly recommended movies

Example Value Model

```
{
  "recommendations": [
    12
  ]
}
```

500

Internal server error

Code

Description

Example Value

Model

```
{  
  "error": "Failed to collect movie data"  
}
```

GET

/recommendations/friends/{user_id}/

Recommend movies watched by friends

get_recommendations_friends__user_id__

Parameters

Try it out

Name

Description

user_id * required

integer (path)

ID of the user requesting recommendations

user_id - ID of the user requesting recommen

Responses

Response content type

application/json

Code

Description

200

List of recommended movies

Example Value

Model

```
{  
  "recommendations": [  
    1  
  ]  
}
```

404

User not found

Example Value

Model

```
{  
  "error": "User with ID 999 not found"  
}
```

500

Internal server error

Example Value

Model

CodeDescription

```
{
  "error": "Failed to fetch watch history"
}
```

GET/recommendations/top/{user_id}/Get top rated moviesget_recommendations_top__user_id__

Parameters

Try it out

NameDescription

user_id * required

integer (path)

ID of the user requesting recommendations

user_id - ID of the user requesting recommen

ResponsesResponse content typeapplication/json

CodeDescription

200

List of top rated movies

Example ValueModel

```
[
  {
    "avg_score": 8.5,
    "movie_id": 1
  }
]
```

404

User not found

Example ValueModel

```
{
  "error": "User with ID 999 not found"
}
```

500

Internal server error

Example ValueModel

localhost:5004/apidocs/#!/Recommendations/get_recommendations_top__user_id__

3/4

Code**Description**

```
{  
  "error": "Failed to fetch ratings"  
}
```

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/apispec_1.json

Explore

Newsfeed Service API 1.0.0

[Base URL: localhost:5005/]
[/apispec_1.json](#)

API for user newsfeed

Schemes

HTTP

Newsfeed



GET

/newsfeed/{user_id}/

Get newsfeed of recent movies watched by friends

get_newsfeed__user_id__

Parameters

Try it out

Name	Description
user_id * required integer (path)	ID of the user requesting the newsfeed

user_id - ID of the user requesting the newsfeed

Responses

Response content type

application/json

Code	Description
200	List of recent movies watched by friends

Example Value

Model

```
{  "newsfeed": [    {      "friend_id": 2,      "friend_username": "janedoe",      "movie_id": 1,
```

Code	Description
	<pre>"timestamp": "2025-05-01T10:00:00Z" }] }</pre>
404	User not found Example Value Model <pre>{ "error": "User with ID 999 not found" }</pre>
500	Internal server error Example Value Model <pre>{ "error": "Failed to fetch friends" }</pre>

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/apispec_1.json

Explore

Movie Service API 1.0.0

[Base URL: localhost:5006/]
[/apispec_1.json](#)

API for movie information

Schemes

HTTP

Movies



GET

/movie/{id}/ Get movie by ID

get_movie_id__

Parameters

Try it out

Name	Description
id * required	
integer (path)	ID of the movie
id - ID of the movie	

Responses

Response content type application/json

Code	Description
200	Movie details
Example Value Model { "Genre": "Drama", "ID": 1, "Metascore": 80, "Movie Name": "The Shawshank Redemption", "Plot": "Two imprisoned men bond over a number of years, finding solace and	

Code

Description

```
eventual redemption through acts of common decency.",
  "Rating": 9.3,
  "Runtime": "142 min"
}
```

404

Movie not found

Example Value Model

```
{
  "error": "Movie with ID 999 not found"
}
```

GET

/movies/ Get a list of movies

get_movies_

Parameters

Try it out

Name

Description

start
integer
(query)

Starting index (0-based)

Example : 0

start - Starting index (0-based)

count
integer
(query)

Number of movies to return

Example : 50

count - Number of movies to return

Responses

Response content type

application/json

Code

Description

200

List of movies

Example Value Model

```
[
  {
    "Genre": "Drama",
    "ID": 1,
```

Code	Description
	<pre>"Metascore": 80, "Movie Name": "The Shawshank Redemption", "Plot": "Two imprisoned men bond over a number of years, finding solace and eventual redemption through acts of common decency.", "Rating": 9.3, "Runtime": "142 min" }]</pre>

400

Invalid parameters

Example Value Model

```
{  
  "error": "Start parameter cannot be negative"  
}
```

404

No movies found

Example Value Model

```
{  
  "error": "Start parameter exceeds dataset size"  
}
```

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